

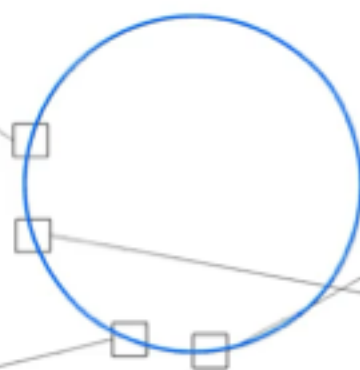
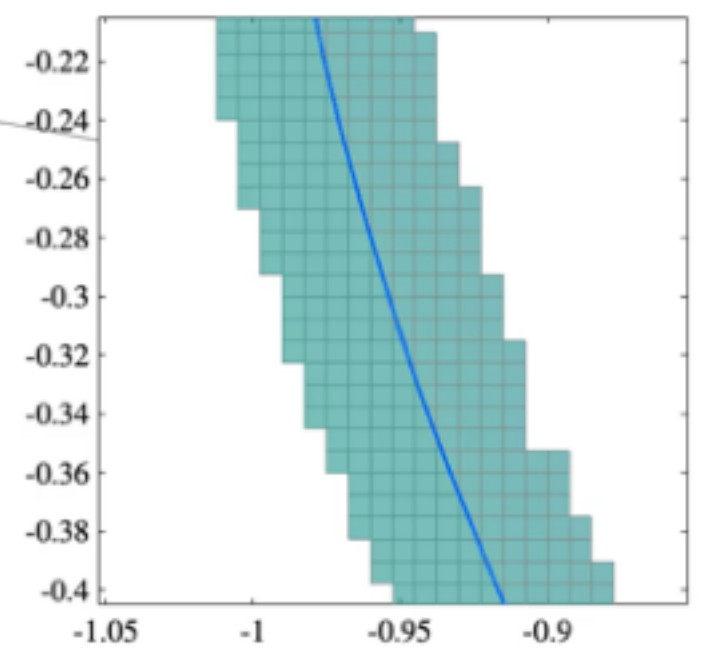
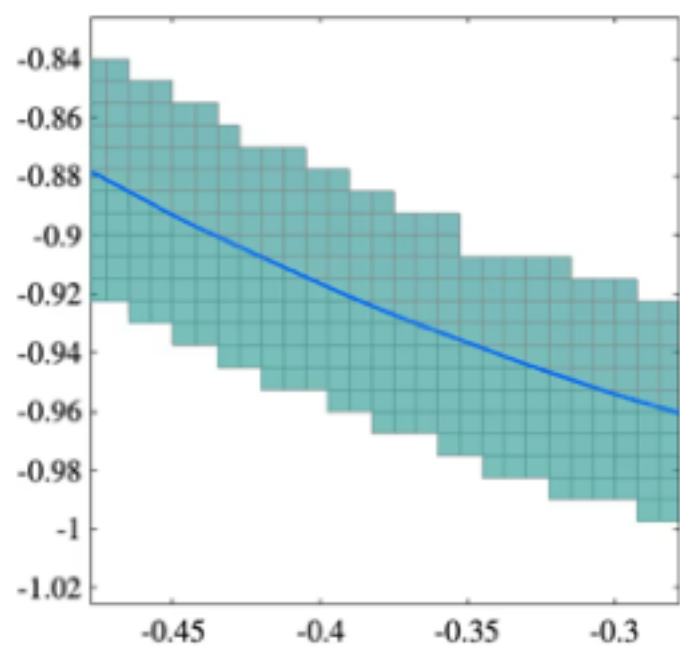
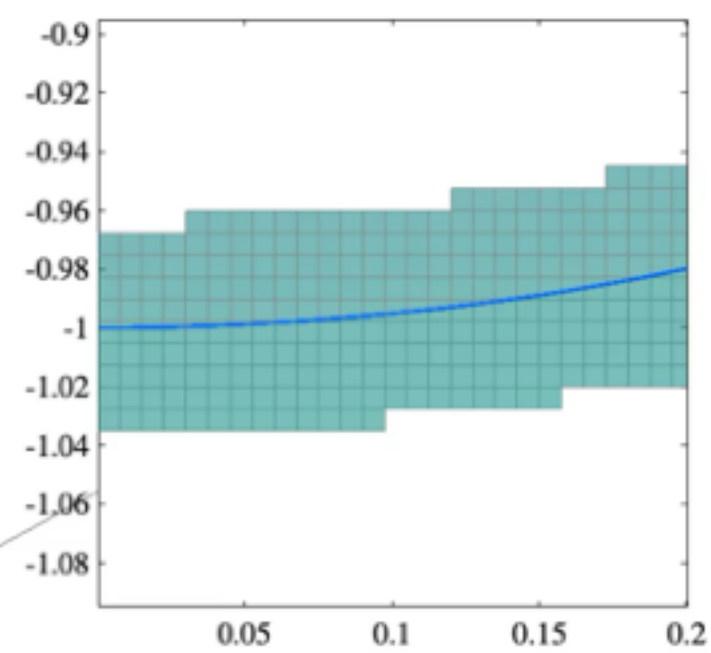
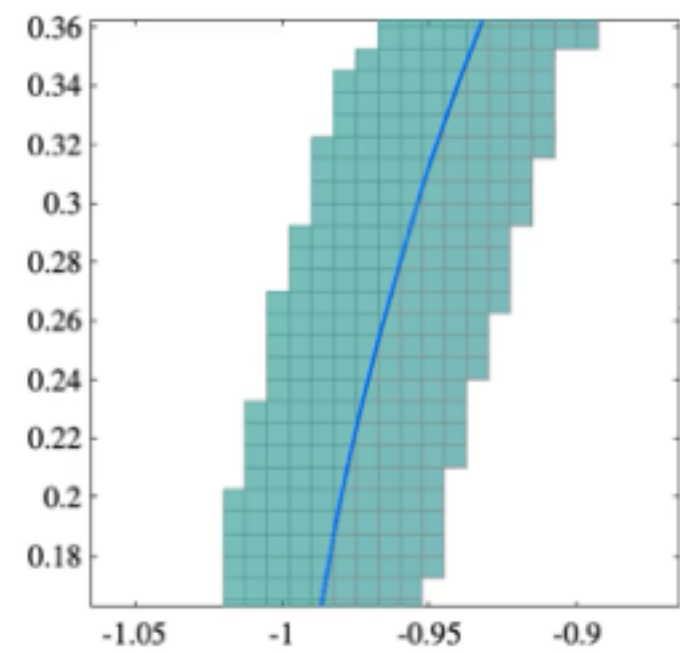


# MICROSCALE FILAMENTATION BY ALTERNATING SHEARS





$h/L = 1.25\text{e-}03$ ,  $t = 0$ ,  $N(t) = 1$



# TOPOLOGICAL RIGIDITY

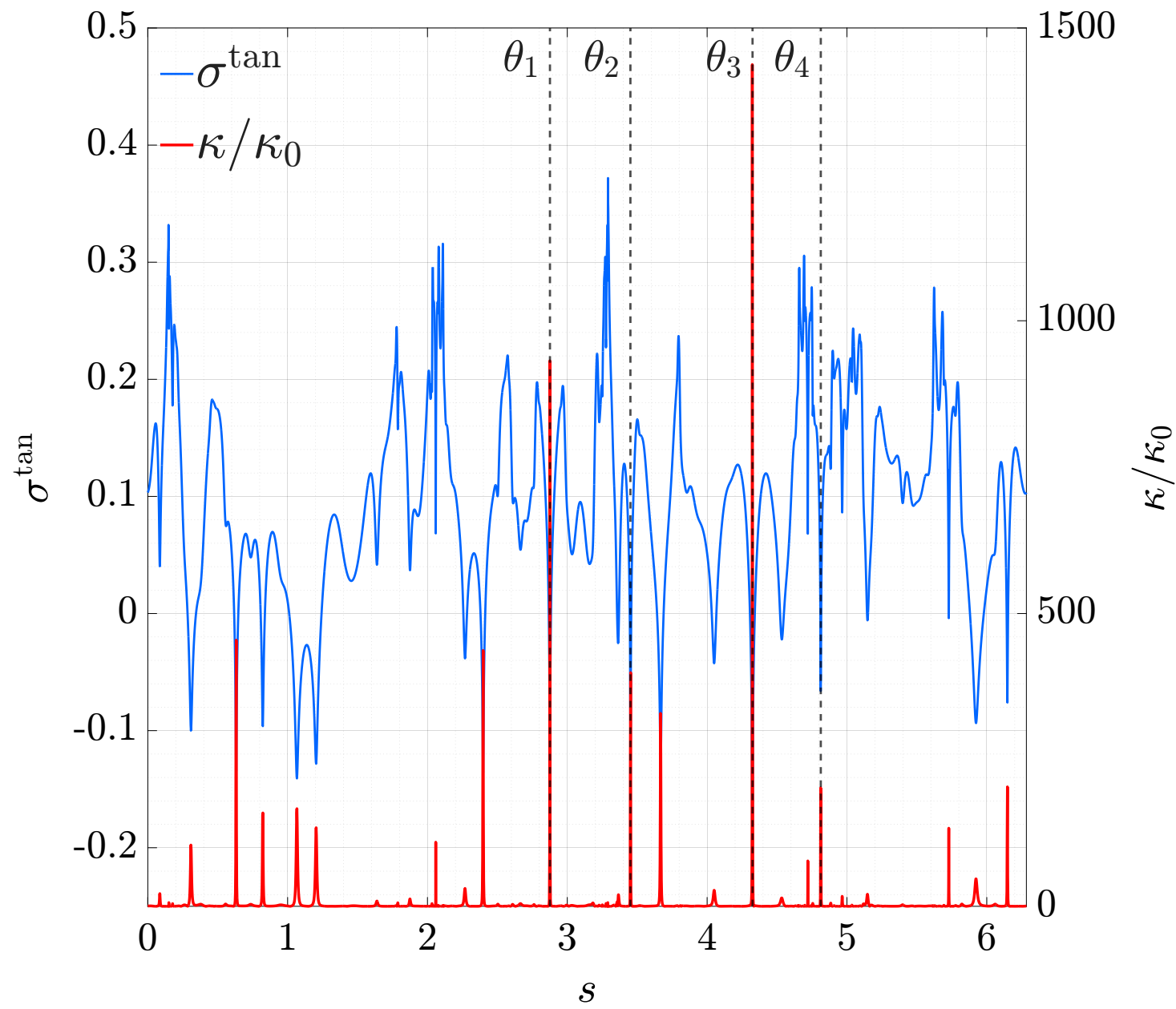
Filaments collapse below the microscale

$$\frac{h}{L} = \mathcal{O}(10^{-3})$$

# MULTISCALE CHAOTIC FLOW

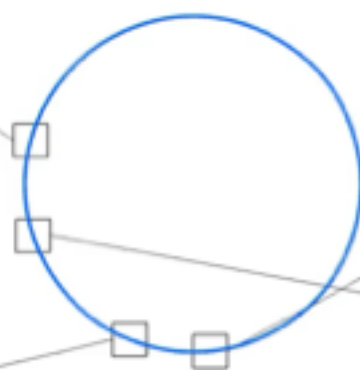
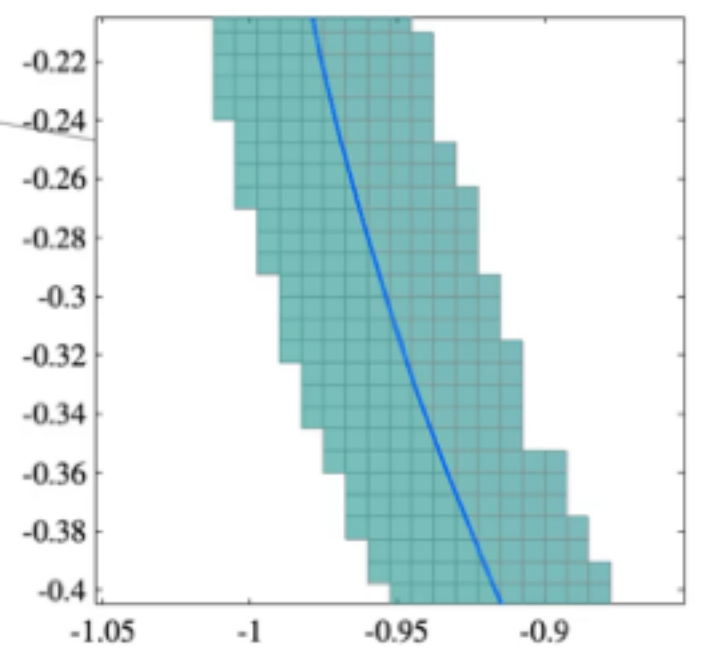
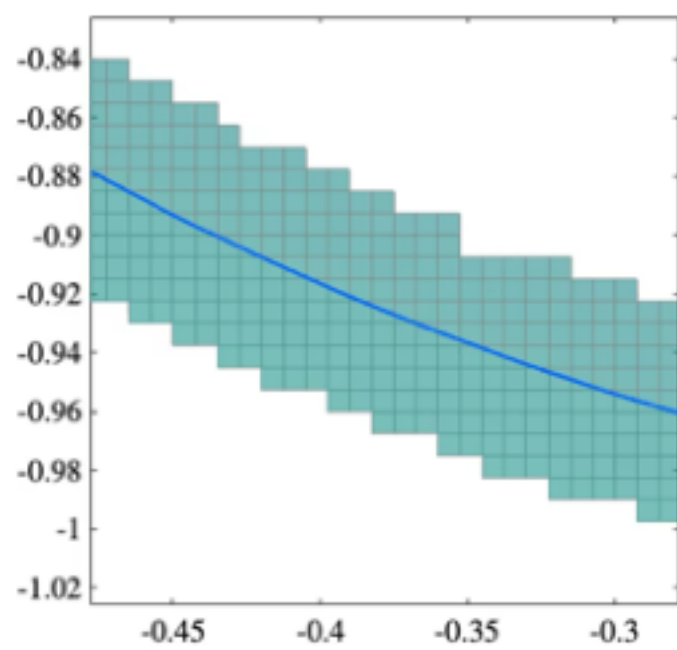
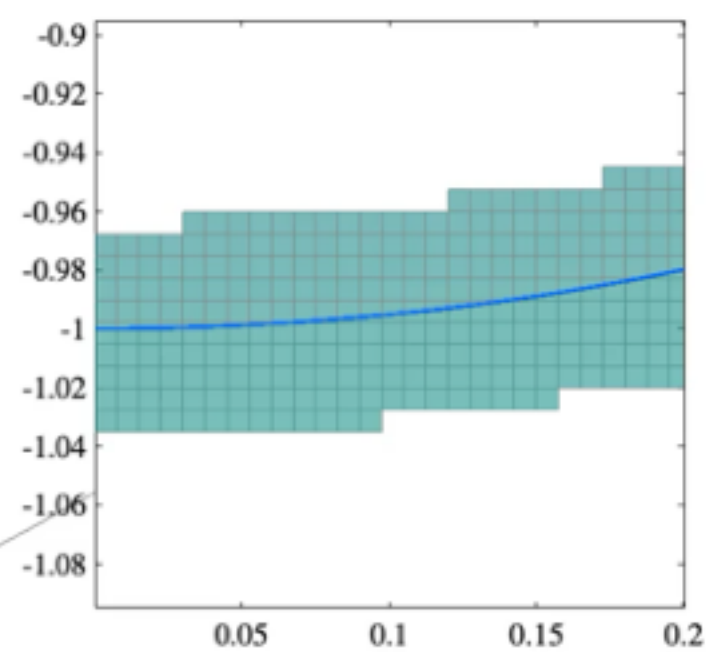
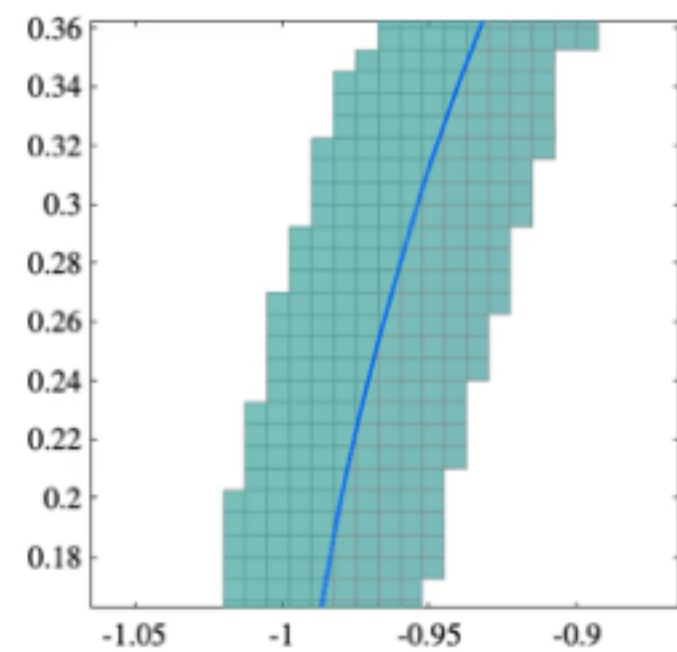
Alternating shears generate exponential stretching and transverse compression.

Geometric & material diagnostics predict where filaments emerge.

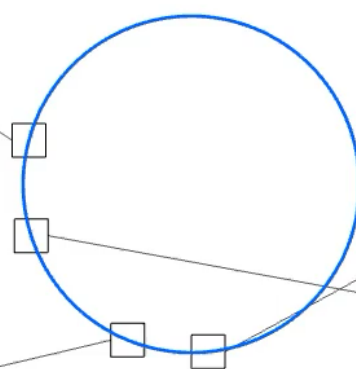
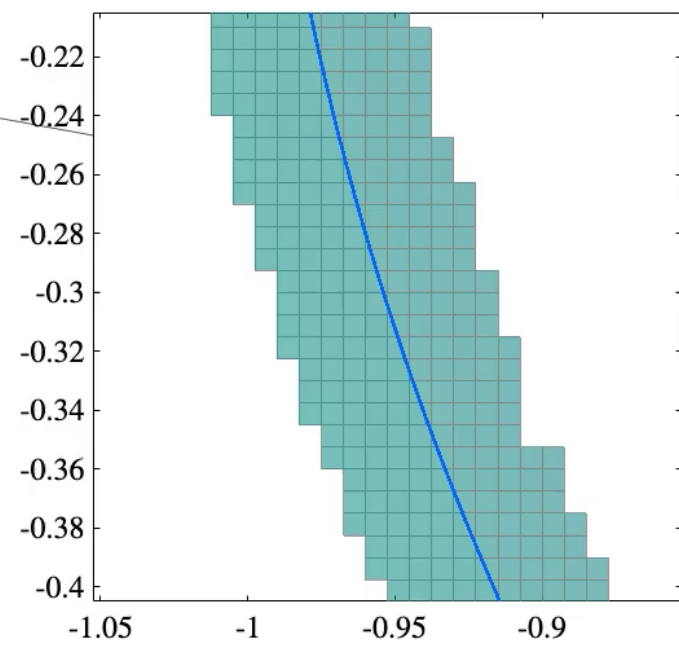
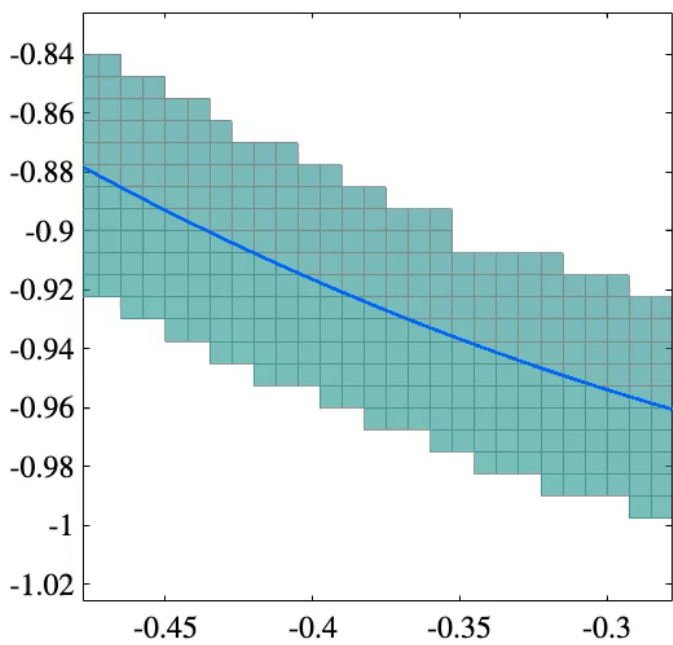
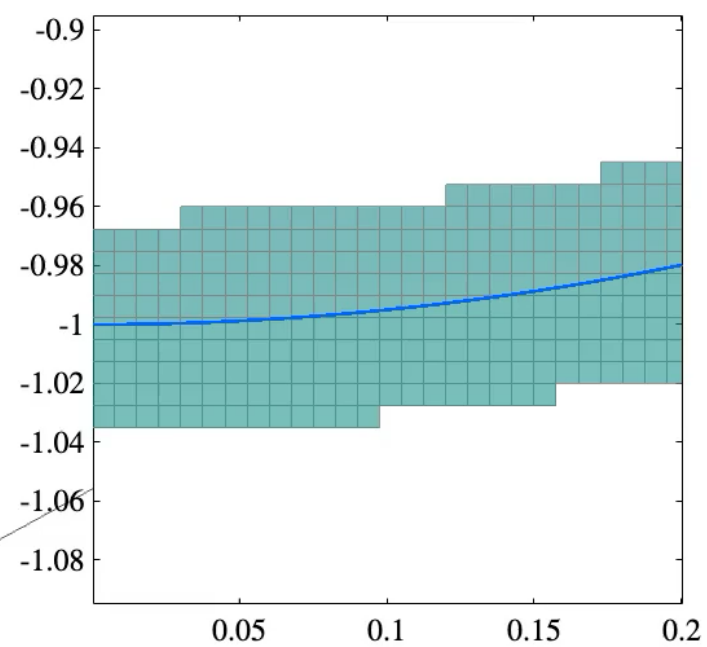
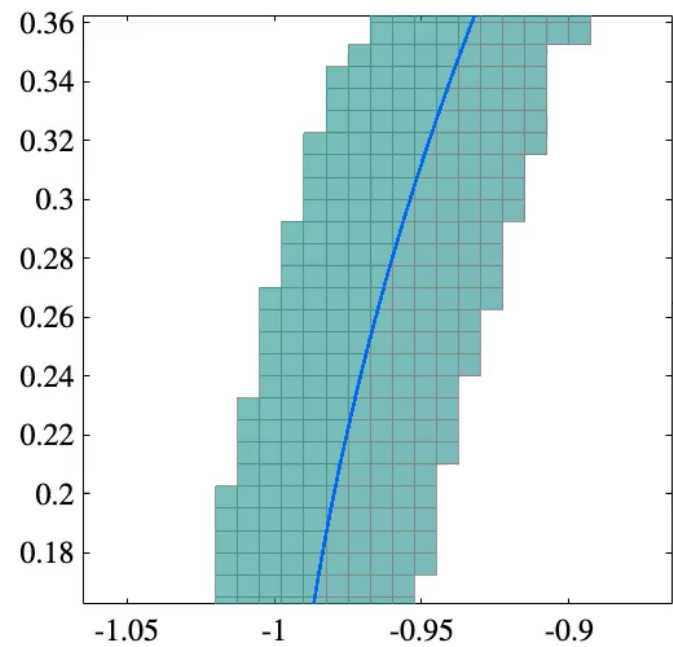




$h/L = 1.25\text{e-}03$ ,  $t = 0$ ,  $N(t) = 1$

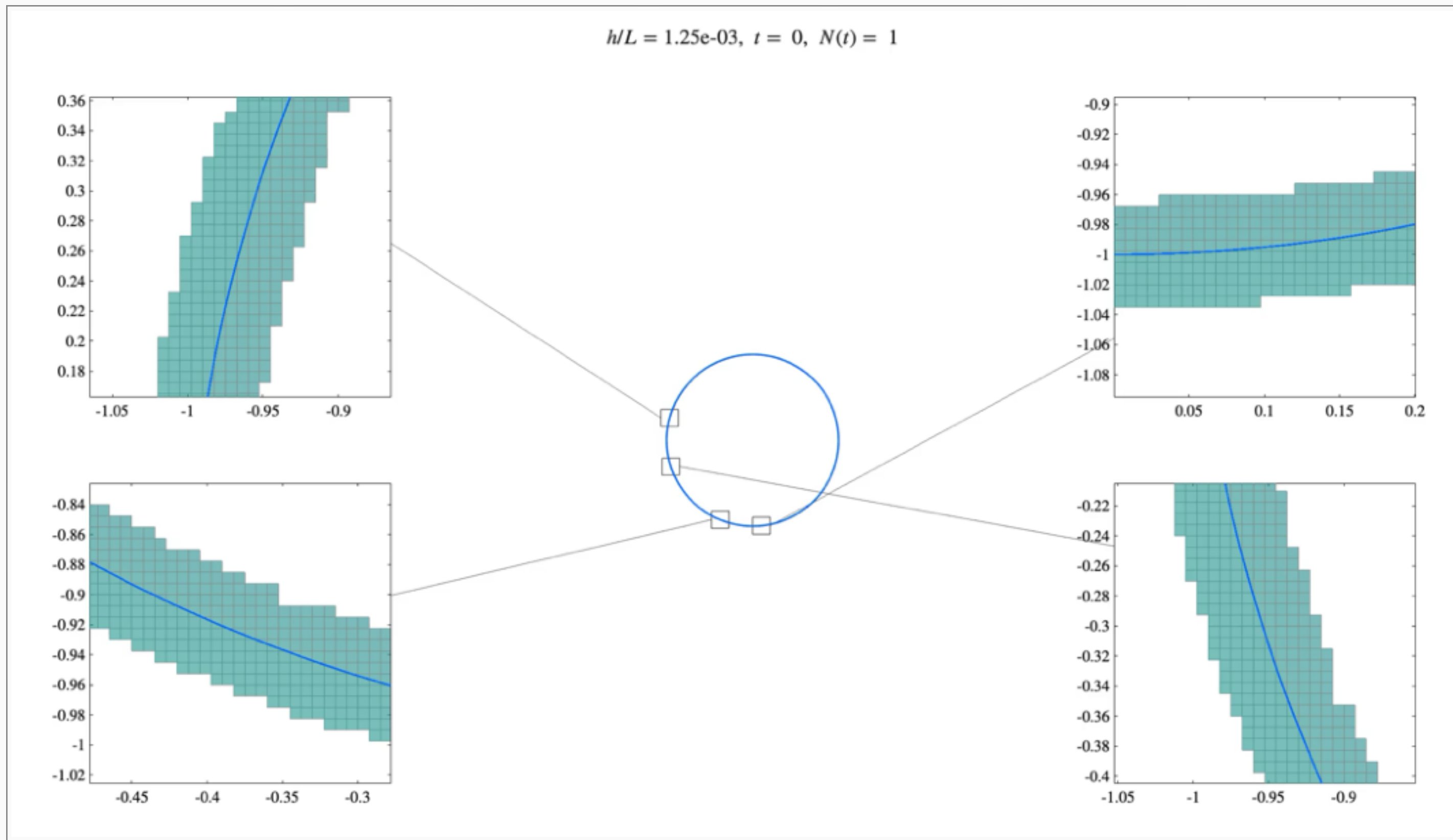


$$h/L = 1.25\text{e-}03, \quad t = 0, \quad N(t) = 1$$





# MICROSCALE FILAMENTATION BY ALTERNATING SHEARS



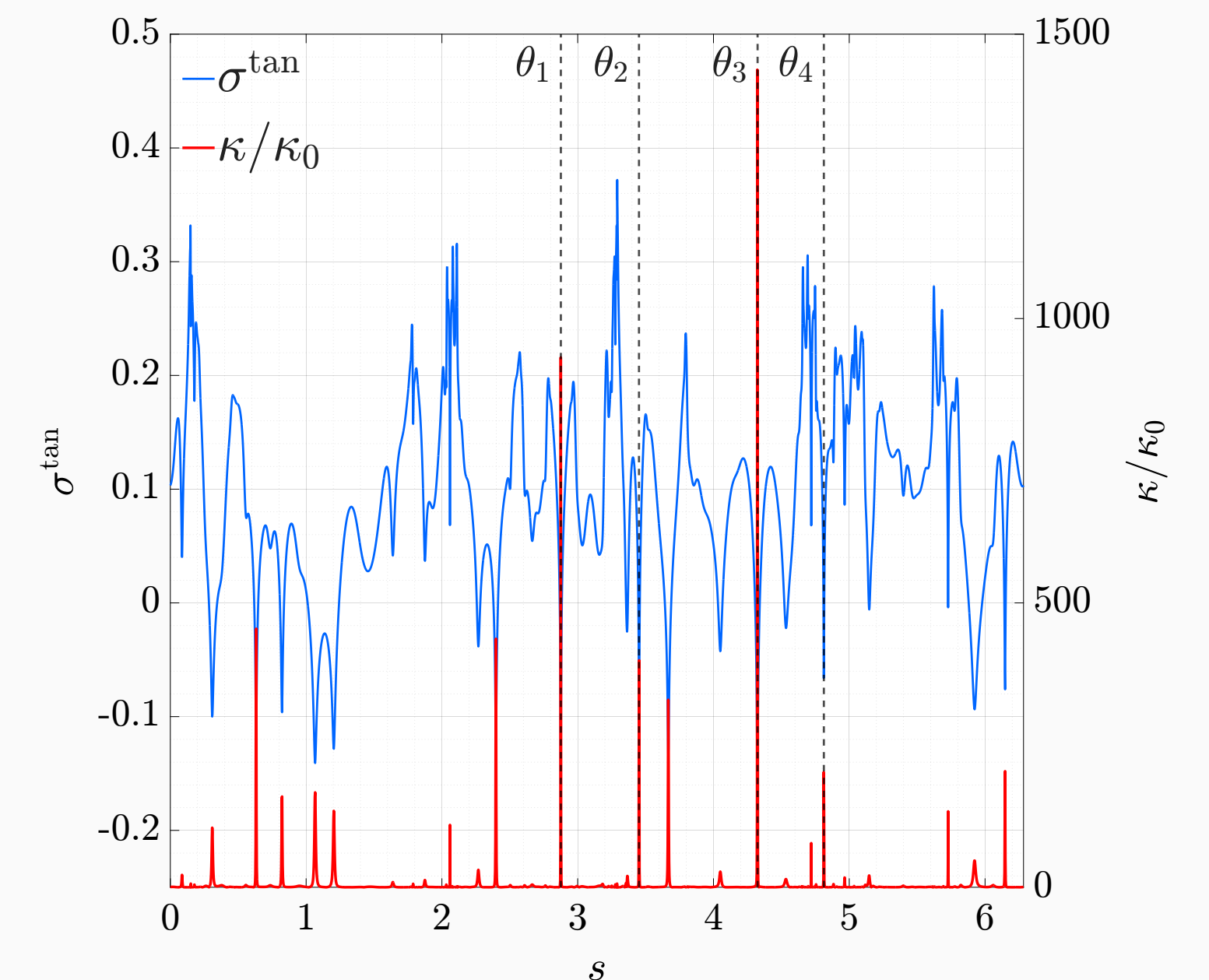
## TOPOLOGICAL RIGIDITY

Filaments collapse below the microscale  $\frac{h}{L} = \mathcal{O}(10^{-3})$

## MULTISCALE CHAOTIC FLOW

Alternating shears generate exponential stretching and transverse compression.

Geometric & material diagnostics predict where filaments emerge.





## Quasi-incompressible Cahn–Hilliard fluids and topological transitions

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### DIFFUSE-INTERFACE REGULARIZATION

Filament collapse is regularized

by a diffuse layer of width  $\epsilon$

